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PF Lab class activity (Arrays)

1. Store 10 integers in array and print them

```
#include <iostream> Untitled-1
#include <iostream>
using namespace std;

int main() {
    int arr[10], num;
    bool found = false;

    cout << "Enter 10 integers:\n";
    for(int i = 0; i < 10; i++) {
        cin >> arr[i];
    }

    cout << "Enter number to search: ";
    cin >> num;

    for(int i = 0; i < 10; i++) {
        if(arr[i] == num) {
            found = true;
            break;
        }
    }

    if(found)
        cout << "Number found";
    else
        cout << "Number not found";

    return 0;
}
```

2. Check whether a number exists in array

```
#include <iostream> Untitled-1
#include <iostream>
using namespace std;

int main() {
    int arr[10];

    cout << "Enter 10 integers:\n";
    for(int i = 0; i < 10; i++) {
        cin >> arr[i];
    }

    cout << "Elements are:\n";
    for(int i = 0; i < 10; i++) {
        cout << arr[i] << " ";
    }

    return 0;
}
```

3. Count positive, negative, odd, even and zeros

```
#include <iostream>
using namespace std;

int main() {
    int arr[20];
    int positive = 0, negative = 0, odd = 0, even = 0, zero = 0;

    cout << "Enter 20 integers:\n";

    for(int i = 0; i < 20; i++) {
        cin >> arr[i];

        if(arr[i] > 0)
            positive++;

        else if(arr[i] < 0)
            negative++;

        else
            zero++;

        if(arr[i] % 2 == 0)
            even++;
        else
            odd++;
    }

    cout << "Positive: " << positive << endl;
    cout << "Negative: " << negative << endl;
    cout << "Odd: " << odd << endl;
    cout << "Even: " << even << endl;
    cout << "Zeros: " << zero << endl;

    return 0;
}
```

4. Copy array in reverse order

```cpp
#include <iostream>
using namespace std;

int main() {
    int arr[10], rev[10];

    cout << "Enter 10 integers:\n";

    for(int i = 0; i < 10; i++) {
        cin >> arr[i];
    }

    for(int i = 0; i < 10; i++) {
        rev[i] = arr[9 - i];
    }

    cout << "Reversed array:\n";

    for(int i = 0; i < 10; i++) {
        cout << rev[i] << " ";
    }

    return 0;
}
```

5. Find sum and product of array elements

```cpp
#include <iostream>
using namespace std;

int main() {
    int arr[5];
    int sum = 0, product = 1;

    cout << "Enter 5 elements:\n";

    for(int i = 0; i < 5; i++) {
        cin >> arr[i];
        sum += arr[i];
        product *= arr[i];
    }

    cout << "Sum = " << sum << endl;
    cout << "Product = " << product << endl;

    return 0;
}
```

6. Find largest and smallest element

```cpp
#include <iostream>
using namespace std;

int main() {
    int arr[5];

    cout << "Enter 5 elements:\n";

    for(int i = 0; i < 5; i++) {
        cin >> arr[i];
    }

    int largest = arr[0];
    int smallest = arr[0];

    for(int i = 1; i < 5; i++) {
        if(arr[i] > largest)
            largest = arr[i];

        if(arr[i] < smallest)
            smallest = arr[i];
    }

    cout << "Largest = " << largest << endl;
    cout << "Smallest = " << smallest << endl

    return 0;
}
```

7. Check palindrome array

```cpp
#include <iostream>
using namespace std;

int main() {
    int arr[8];
    bool palindrome = true;

    cout << "Enter 8 elements:\n";

    for(int i = 0; i < 8; i++) {
        cin >> arr[i];
    }

    for(int i = 0; i < 4; i++) {
        if(arr[i] != arr[7 - i]) {
            palindrome = false;
            break;
        }
    }

    if(palindrome)
        cout << "Palindrome";
    else
        cout << "Not Palindrome";

    return 0;
}
```

8. Split array into two arrays

```cpp
#include <iostream>
using namespace std;

int main() {
    int arr[10], arr1[5], arr2[5];

    cout << "Enter 10 elements:\n";

    for(int i = 0; i < 10; i++) {
        cin >> arr[i];
    }

    for(int i = 0; i < 5; i++) {
        arr1[i] = arr[i];
    }

    for(int i = 5; i < 10; i++) {
        arr2[i - 5] = arr[i];
    }

    cout << "First array:\n";
    for(int i = 0; i < 5; i++) {
        cout << arr1[i] << " ";
    }

    cout << "\nSecond array:\n";
    for(int i = 0; i < 5; i++) {
        cout << arr2[i] << " ";
    }

    return 0;
}
```

9. Find pair with maximum and minimum difference

```cpp
#include <iostream>
using namespace std;

int main() {
    int n;

    cout << "Enter size of array: ";
    cin >> n;

    int arr[n];

    cout << "Enter elements:\n";

    for(int i = 0; i < n; i++) {
        cin >> arr[i];
    }

    int maxDiff = arr[1] - arr[0];
    int minDiff = arr[1] - arr[0];

    int maxA = arr[0], maxB = arr[1];
    int minA = arr[0], minB = arr[1];

    for(int i = 0; i < n; i++) {
        for(int j = i + 1; j < n; j++) {

            int diff = abs(arr[i] - arr[j]);

            if(diff > maxDiff) {
                maxDiff = diff;
                maxA = arr[i];
                maxB = arr[j];
            }

            if(diff < minDiff) {
                minDiff = diff;
                minA = arr[i];
                minB = arr[j];
            }
        }
    }

    cout << "Maximum difference pair: "
         << maxA << " and " << maxB << endl;

    cout << "Minimum difference pair: "
         << minA << " and " << minB << endl;

    return 0;
}
```

10. Find subarray between indices 3 and 8

```cpp
#include <iostream>
using namespace std;

int main() {

    int arr[] = {10, 12, 20, 30, 25, 40, 32, 31, 35, 50, 60};

    cout << "Subarray is:\n";

    for(int i = 3; i <= 8; i++) {
        cout << arr[i] << " ";
    }

    return 0;
}
```

11. Print sum, average, smallest and largest

```cpp
#include <iostream>
using namespace std;

int main() {
    int arr[5];
    int sum = 0;

    cout << "Enter 5 elements:\n";

    for(int i = 0; i < 5; i++) {
        cin >> arr[i];
        sum += arr[i];
    }

    float average = sum / 5.0;

    int largest = arr[0];
    int smallest = arr[0];

    for(int i = 1; i < 5; i++) {

        if(arr[i] > largest)
            largest = arr[i];

        if(arr[i] < smallest)
            smallest = arr[i];
    }

    cout << "Sum = " << sum << endl;
    cout << "Average = " << average << endl;
    cout << "Largest = " << largest << endl;
    cout << "Smallest = " << smallest << endl;

    return 0;
}
```

12. Circular right shift

```cpp
#include <iostream>
using namespace std;

int main() {

    int arr[5];

    cout << "Enter 5 elements:\n";

    for(int i = 0; i < 5; i++) {
        cin >> arr[i];
    }

    int last = arr[4];

    for(int i = 4; i > 0; i--) {
        arr[i] = arr[i - 1];
    }

    arr[0] = last;

    cout << "After circular shift:\n";

    for(int i = 0; i < 5; i++) {
        cout << arr[i] << " ";
    }

    return 0;
}
```